A LIOUVILLE THEOREM FOR STATIONARY NAVIER–STOKES IN $L^{3,\infty}(\mathbb{R}^3)$, AND THE ENDPOINT GAP IN THE CLAY REGULARITY CHAIN

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Abstract. We prove that every smooth stationary divergence-free solution of the incompressible three-dimensional Navier–Stokes equations that lies in the Lorentz space $L^{3,\infty}(\mathbb{R}^3)$ is identically zero. The proof is entirely first-principles: a hat-function weighted energy on fixed-ratio annuli is combined with exact pressure elimination via the Bogovskiĭ operator, a Lorentz layer-cake embedding, and a rescaled Gagliardo–Nirenberg interpolation. Exact ρ-cancellation at scaling exponent $3a/p + b = 3$ yields a superlinear differential inequality $\mathcal{E}'(\rho) \geq c\rho^{-\alpha}\mathcal{E}(\rho)^{\beta}$ whose exponents satisfy $\beta > 1$, $\alpha < 1$ for every $p \in (3/2, 3)$; integration against bounded left-hand side produces a contradiction that forces finite Dirichlet energy, which a Galdi-style testing argument reduces to $v = 0$.

The theorem completes the Koch–Nadirashvili–Seregin–Šverák (KNSS) program for the general (non-axisymmetric) three-dimensional case. Applied within the KNSS blow-up framework, it reduces the full Clay Millennium regularity problem to a single structural inheritance step: the rescaled ancient solution's uniform $L^{3,\infty}$ bound. That step is equivalent to Phuc's 2015 open endpoint, which remains unresolved for non-axisymmetric flows. We give an honest account of this gap, survey the partial closures (axisymmetric, smallness, finite singular set, log-improvement), and describe a concrete weighted-moment proposal. The companion corpus at `https://github.com/hirunation/finish_navier_stokes` (archived at Zenodo, DOI 10.5281/zenodo.19720003) contains the full provenance-tracked knowledge graph, nine canonical obstruction theorems, thirty-five catalogued dead ends, and the Lean 4 formalization.

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1. Introduction

1.1. **The problem.** For the incompressible Navier–Stokes equations in $\mathbb{R}^3$,

$$\partial_t u - \nu\,\Delta u + (u\cdot\nabla)u + \nabla p = 0, \qquad \nabla\cdot u = 0, \qquad u(\cdot,0) = u_0, \tag{1}$$

the Clay Millennium Prize problem asks whether every smooth Schwartz-class divergence-free u_0 admits a global smooth solution. Leray [16] proved existence of weak solutions in 1934; regularity for general smooth data remains open.

The Koch–Nadirashvili–Seregin–Šverák blow-up framework [14] reduces the regularity question to a Liouville-type theorem. If a smooth solution develops a finite-time singularity at T^*, a rescaling procedure ([14, §6]) extracts a bounded smooth ancient solution $v\colon \mathbb{R}^3\times(-\infty,0]\to\mathbb{R}^3$ with $|v(0,0)| = 1$. The backward-time limit of this ancient solution is a smooth stationary divergence-free Navier–Stokes solution; if that limit were forced to be identically zero, Escauriaza–Seregin–Šverák backward uniqueness [9] would contradict the normalization $|v(0,0)| = 1$, closing the regularity problem.

KNSS [14, §5] established such a Liouville theorem under axial symmetry, using stream-function arguments that reduce 3D to 2D. For general (non-axisymmetric) three-dimensional flows their method does not extend, and they flag this as an open problem. The present paper supplies the general-case Liouville theorem under the critical Lorentz-space hypothesis $v \in L^{3,\infty}(\mathbb{R}^3)$.

1.2. **Main theorem.**

Theorem 1.1 (Stationary $L^{3,\infty}$ Liouville)**.** *Let $v\colon \mathbb{R}^3 \to \mathbb{R}^3$ be smooth, divergence-free, and satisfy the stationary Navier–Stokes equations*

$$-\nu\,\Delta v + (v\cdot\nabla)v + \nabla p = 0, \qquad \nabla\cdot v = 0 \quad \text{on } \mathbb{R}^3,$$

with $\|v\|_{L^{3,\infty}(\mathbb{R}^3)} < \infty$. Then $v \equiv 0$.

Theorem 1.1 completes the KNSS program for the general three-dimensional case. Its proof is self-contained and uses only first-principles tools (the Bogovskiĭ operator for pressure elimination, Lorentz layer-cake embedding, Gagliardo–Nirenberg interpolation); no ancient-Liouville conjectures are invoked.

The critical exponents at the natural choice $p = 2$ are $\beta = 4/3$, $\alpha = 2/3$, and $\gamma = -1/4$. The scale exponent γ is defined in Theorem 7.3 (§7), and the ODE exponents α and β in Proposition 8.1 (§8); Sections 4–10 carry the full integrated argument.

1.3. **The Clay chain and the $L^{3,\infty}$ inheritance gap.** Applied within the KNSS framework, Theorem 1.1 reduces the Clay regularity question to a single structural step. The rescaled ancient solution produced by [14, §6] is bounded in L^∞; what is required in addition, for Theorem 1.1 to apply to its backward-time limit, is a uniform $L^{3,\infty}$ bound on the ancient solution. Scale invariance reduces this question to the spatial decay of the original solution in the exterior of the hypothetical singular set, uniformly on $[0, T^*)$.

This inheritance step is currently open. It is equivalent in scope to Phuc's 2015 open problem on the $L^{3,\infty}$ endpoint regularity criterion [22, p. 4]:

> "To the best of our knowledge, a criterion of local regularity for the Navier–Stokes equations in $L^\infty(-1, 0; L^{3,\infty}(B_1))$ is still unknown."

The obstruction is structural: the Lorentz quasi-norm $\|\cdot\|_{L^{3,\infty}(\mathbb{R}^3)}$ is not absolutely continuous, so exterior-tail compactness arguments that close the $L^{3,q}$ endpoint for $q < \infty$ fail at $q = \infty$. Partial closures exist: the axisymmetric case (Ożański–Palasek [20] via Nazarov–Ural'tseva Harnack [19] on rv_θ), the smallness regime (Luo–Tsai [17], Kim–Kozono [13]), and partial regularity in the sense of finite singular sets (Seregin [24]).

Section 12 describes these partial closures and gives a concrete proposal (§12.4) for closing the general case under the weighted hypothesis $(1 + |x|)u_0 \in L^\infty$, which is automatic for Schwartz data. A full proof of that proposal is outside the scope of the present paper.

1.4. **Corollaries.**

Corollary 1.2 (Four-dimensional Liouville)**.** *Let $v\colon \mathbb{R}^4 \to \mathbb{R}^4$ be smooth, divergence-free, and satisfy the stationary Navier–Stokes equations on $\mathbb{R}^4$ with $v \in L^{4,\infty}(\mathbb{R}^4)$. Then $v \equiv 0$.*

Corollary 1.2 follows from a four-dimensional specialization of the proof at $p = 5/2$, where the dimensional discriminant $\gamma(n, p) = -(n - 4)(n - 2p)/(2n - p(n - 2))$ vanishes identically, producing the borderline logarithmic divergence $\int \rho^{-1}\, d\rho$ against a bounded left-hand side. The discriminant becomes positive for $n \geq 5$, and the method provably fails there: $\beta > 1$ requires $p > n/2$ (Sobolev) while $\alpha \leq 1$ requires $p \leq n/2$ (integration), an incompatibility.

Corollary 1.3 (Weak-solution extension)**.** *The conclusion of Theorem 1.1 holds for $v \in L^{3,\infty}(\mathbb{R}^3) \cap W^{1,2}_{\mathrm{loc}}(\mathbb{R}^3)$.*

The $W^{1,2}_{\mathrm{loc}}$ threshold is sharp: Landau solutions $v(x) = c/|x|$ are smooth outside the origin, lie exactly in $L^{3,\infty}(\mathbb{R}^3)$, and belong to $W^{1,2}_{\mathrm{loc}}$ away from the origin but not at the origin. They realize the boundary case.

Corollary 1.4 (Quantitative bound)**.** *From the superlinear ODE at $p = 2$, the weighted Dirichlet energy on a scale-ρ_0 annulus satisfies the explicit bound $\mathcal{E}(\rho_0) \leq C\, M^6\, \rho_0^{-1}$, with explicit constants.*

1.5. **Companion corpus.** The complete corpus — 111 provenance-tracked smart-data entries spanning insights, lemmas, definitions, obstruction theorems, classical axioms, anti-knowledge entries, attribution corrections, master proof chains, and cross-cutting narrative threads — is maintained publicly at

https://github.com/hirunation/finish_navier_stokes

and archived at Zenodo with DOI 10.5281/zenodo.19720003. The repository includes a Lean 4 skeleton for the main theorem (`formal/NS/INS_01_L3inf_Liouville.lean`), literature reviews for the nine obstruction theorems of §13, the thirty-five catalogued dead ends, and the desk-level attack analyses referenced throughout this paper. Readers seeking the full architecture of the Clay-chain program, the obstruction landscape, or the alternative-route research programs (Hull-Ergodic, Icosahedral, Ancient Extension) are directed there.

1.6. **Organization.** Section 2 fixes notation. Sections 3–10 prove Theorem 1.1 in eight steps. Section 11 proves the Corollaries 1.2–1.4. Section 12 describes the application to the Clay chain and documents the $L^{3,\infty}$ inheritance gap. Section 13 gives a brief summary of the nine-obstruction landscape. Section 14 concludes with discussion and open problems.

2. Notation and preliminaries

Throughout, $\nu > 0$ is a fixed viscosity and v denotes a smooth divergence-free stationary solution of (1) on $\mathbb{R}^3$ with

$$M := \|v\|_{L^{3,\infty}(\mathbb{R}^3)} < \infty.$$

We assume $M > 0$; otherwise $v = 0$ almost everywhere, hence identically by smoothness, and there is nothing to prove.

Notation 2.1 (Geometric objects)**.** Fix $\theta \in (0,1)$. For $\rho > 0$, write

$$B_\rho := \{x \in \mathbb{R}^3 : |x| < \rho\}, \qquad A(\theta\rho, \rho) := B_\rho \setminus \overline{B_{\theta\rho}}.$$

Define the *hat function* $\varphi\colon [0,\infty) \to [0,1]$ by

$$\varphi(t) = \begin{cases} 1 & 0 \le t < \theta, \\ \dfrac{1-t}{1-\theta} & \theta \le t < 1, \\ 0 & t \ge 1. \end{cases}$$

The *weighted energy at scale* ρ is

$$\mathcal{E}(\rho) := \int_{\mathbb{R}^3} |\nabla v(x)|^2 \, \varphi(|x|/\rho) \, dx,$$

and the *annular gradient norm* is

$$D(\rho) := \|\nabla v\|^2_{L^2(A(\theta\rho,\rho))}.$$

The hat function is piecewise linear on $[\theta, 1]$ and zero outside $[0,1]$, smooth on the open pieces with a bounded weak gradient. It is $C^{0,1}$ globally, and for the arguments below it may be replaced by any fixed mollification supported in $A(\theta\rho,\rho)$ with the same qualitative properties; we use the piecewise-linear form throughout for the cleanest derivative formula (cf. Remark 2.3).

Lemma 2.2 (Hat-function derivative formula)**.** *For every $\rho > 0$,*

$$\mathcal{E}'(\rho) = \frac{1}{(1-\theta)\rho^2} \int_{A(\theta\rho,\rho)} |\nabla v(x)|^2 |x| \, dx \ge \frac{\theta}{(1-\theta)\rho} \, D(\rho). \tag{2}$$

In particular, $\mathcal{E}$ is absolutely continuous and non-decreasing in ρ, and

$$D(\rho) \le \frac{1-\theta}{\theta}\rho\,\mathcal{E}'(\rho). \tag{3}$$

Proof. Differentiating under the integral sign (valid by v smooth and φ weakly differentiable with compact support in ρ),

$$\mathcal{E}'(\rho) = \int |\nabla v(x)|^2 \frac{d}{d\rho}\varphi(|x|/\rho)\,dx = \int |\nabla v(x)|^2 \frac{|x|}{(1-\theta)\rho^2}\,\chi_{A(\theta\rho,\rho)}(x)\,dx,$$

using $(d/d\rho)\varphi(|x|/\rho) = (|x|/(1-\theta)\rho^2)\chi_{[\theta\rho,\rho]}(|x|)$ from the piecewise-linear definition. This gives the first equality. For the inequality, bound $|x| \ge \theta\rho$ on the annulus:

$$\mathcal{E}'(\rho) \ge \frac{\theta}{(1-\theta)\rho}\int_{A(\theta\rho,\rho)} |\nabla v|^2\,dx = \frac{\theta}{(1-\theta)\rho}\,D(\rho).$$

□

Remark 2.3 (Three purposes of the hat function)**.** The piecewise-linear hat function serves three distinct roles. (i) It makes $\mathcal{E}(\rho)$ monotone in ρ, so any bound on $\mathcal{E}$ at one scale transfers to smaller scales. (ii) Its derivative (2) directly controls the annular gradient integral, giving the conversion between the integrated energy and a pointwise-in-ρ ODE quantity. (iii) Its gradient is supported on the fixed-ratio annulus $A(\theta\rho,\rho)$, ensuring the Bogovskiĭ constants introduced in §4 are uniform in ρ.

A smooth mollification of φ would require writing the derivative as an average over a transition layer of width ϵ; the argument below goes through with that replacement at the cost of an additional ϵ-bookkeeping. We keep the sharp form for clarity.

3. Hat-function energy and the iteration target

The proof of Theorem 1.1 is a scale-by-scale argument on $\mathcal{E}(\rho)$. The goal (reached in §9) is the differential inequality

$$\mathcal{E}'(\rho) \ge c\,\rho^{-\alpha}\,\mathcal{E}(\rho)^{\beta} \qquad \text{with } \beta > 1 \text{ and } \alpha < 1, \tag{4}$$

for ρ above a fixed threshold $\rho_0(M,\nu)$. Integrating against the boundedness of $\mathcal{E}$ at infinity forces $\int_{\mathbb{R}^3} |\nabla v|^2\,dx < \infty$, from which a Galdi-style testing argument (§10) gives $v \equiv 0$.

The path from (2) to (4) consists of: pressure elimination via Bogovskiĭ (§4), a Caccioppoli-type inequality on annuli (§5), Lorentz layer-cake embeddings (§6), and a rescaled Gagliardo–Nirenberg interpolation (§7). Each section produces a sublinear-in-D bound on a piece of $\mathcal{E}$; combined, they give the key estimate

$$\mathcal{E}(\rho) \le C_0 M\rho^{-1/2}D(\rho)^{1/2} + C_1 M^3\rho^{-1} + C_2 M^a\rho^{\gamma}D(\rho)^b,$$

with explicit exponents $a = 3p/(6-p)$, $b = 3(6-2p)/(2(6-p))$, $\gamma = (3-2p)/(6-p)$ for every $p \in (3/2, 3)$. At $p = 2$ these give $a = 6/5$, $b = 3/4$, $\gamma = -1/4$.

4. Pressure elimination via Bogovskiĭ

The stationary Navier–Stokes system carries a pressure field p which, when tested against a non-divergence-free function, produces a boundary contribution that does not close under scaling. The Bogovskiĭ operator converts any compactly supported cutoff of v into a divergence-free competitor at the cost of an explicit L^q correction on the same support. The pressure integral then vanishes identically, at no further cost.

4.1. **The Bogovskiĭ operator on John domains.** For an open set $\Omega \subset \mathbb{R}^3$ we write $L^q_0(\Omega) := \{f \in L^q(\Omega) : \int_\Omega f\,dx = 0\}$.

Lemma 4.1 (Bogovskiĭ)**.** *Let $\Omega \subset \mathbb{R}^3$ be a bounded John domain with John constant c_J. There is a linear operator*

$$\mathcal{B}_\Omega \colon L^q_0(\Omega) \to W^{1,q}_0(\Omega;\mathbb{R}^3), \qquad 1 < q < \infty,$$

such that, for every $f \in L^q_0(\Omega)$,

- *(B1)* $\operatorname{div} \mathcal{B}_\Omega[f] = f$ *in* Ω*, and* $\mathcal{B}_\Omega[f] \in W^{1,q}_0(\Omega;\mathbb{R}^3)$*;*
- *(B2)* $\|\mathcal{B}_\Omega[f]\|_{L^q(\Omega)} + \|\nabla\mathcal{B}_\Omega[f]\|_{L^q(\Omega)} \le C_{q,c_J}\,\|f\|_{L^q(\Omega)}$*;*
- *(B3) if* $f \in L^{q_1}_0(\Omega) \cap L^{q_2}_0(\Omega)$ *with* $1 < q_1, q_2 < \infty$*, then the two definitions agree;*
- *(B4)* $\mathcal{B}_\Omega[f]$ *depends linearly and continuously on* f *with respect to every admissible* q*.*

The construction and the bounds are due to Bogovskiĭ [1] on star-shaped domains; the extension to John domains (in particular, to finite unions of star-shaped sets and to annuli) is standard, see Galdi [10, §III.3].

Fixed-ratio annuli are uniformly John.

Lemma 4.2 (Uniform John constant for annuli)**.** *For every $\theta \in (0,1)$ there is a constant $c_J(\theta)$ such that, for every $\rho > 0$, the annulus $A(\theta\rho,\rho)$ is John with constant $c_J(\theta)$.*

This is elementary: $A(\theta\rho,\rho)$ is the image of $A(\theta,1)$ under the dilation $y \mapsto \rho y$, and the John condition is scale-invariant. We will use this in the following rescaled form.

Proposition 4.3 (Scaling of $\mathcal{B}$ on annuli)**.** *Let $\Omega_1 = A(\theta,1)$ and $\Omega_\rho = A(\theta\rho,\rho)$, let $f \in L^q_0(\Omega_\rho)$, and let $\tilde f(y) := f(\rho y)$. Then $\tilde f \in L^q_0(\Omega_1)$, and*

$$\mathcal{B}_{\Omega_\rho}[f](x) = \rho\,\mathcal{B}_{\Omega_1}[\tilde f](x/\rho), \qquad \nabla\mathcal{B}_{\Omega_\rho}[f](x) = (\nabla_y\mathcal{B}_{\Omega_1}[\tilde f])(x/\rho).$$

In particular,

$$\|\mathcal{B}_{\Omega_\rho}[f]\|_{L^q(\Omega_\rho)} = \rho^{1+3/q}\|\mathcal{B}_{\Omega_1}[\tilde f]\|_{L^q(\Omega_1)}, \quad \|\nabla\mathcal{B}_{\Omega_\rho}[f]\|_{L^q(\Omega_\rho)} = \rho^{3/q}\|\nabla_y\mathcal{B}_{\Omega_1}[\tilde f]\|_{L^q(\Omega_1)}. \tag{5}$$

Proof. The change of variable $y = x/\rho$ in Bogovskiĭ's explicit singular-integral representation (which is a convolution with a kernel of degree $-(n-1)$ in $\mathbb{R}^n$) shows that $\rho\mathcal{B}_{\Omega_1}[\tilde f](\cdot/\rho)$ satisfies the defining properties (B1)–(B4) on Ω_ρ with the prescribed operator norm. The uniqueness of the Bogovskiĭ operator modulo test-function density [10, III.3.1] gives the claimed identity. The norm formulas follow by substitution: $\|\mathcal{B}_{\Omega_\rho}[f]\|^q_{L^q(\Omega_\rho)} = \int_{\Omega_\rho}\rho^q|\mathcal{B}_{\Omega_1}[\tilde f](x/\rho)|^q dx = \rho^{q+3}\int_{\Omega_1}|\mathcal{B}_{\Omega_1}[\tilde f](y)|^q dy$. □

Combining Lemmas 4.1–4.2 and Proposition 4.3 gives the uniform-in-ρ norm bound that will be used throughout.

Corollary 4.4 (Uniform annular Bogovskiĭ bound)**.** *For every $\theta \in (0,1)$ and every $q \in (1,\infty)$ there is a constant $C_{q,\theta}$ such that, for every $\rho > 0$ and every $f \in L^q_0(A(\theta\rho,\rho))$,*

$$\|\mathcal{B}_{A(\theta\rho,\rho)}[f]\|_{L^q} \le C_{q,\theta}\,\rho\,\|f\|_{L^q}, \qquad \|\nabla\mathcal{B}_{A(\theta\rho,\rho)}[f]\|_{L^q} \le C_{q,\theta}\,\|f\|_{L^q}. \tag{6}$$

Proof. From (5), $\|\mathcal{B}_{\Omega_\rho}[f]\|_{L^q} = \rho^{1+3/q}\|\mathcal{B}_{\Omega_1}[\tilde f]\|_{L^q} \le C_{q,\theta}\rho^{1+3/q}\|\tilde f\|_{L^q} = C_{q,\theta}\rho\cdot\rho^{3/q}\|\tilde f\|_{L^q(\Omega_1)} = C_{q,\theta}\rho\|f\|_{L^q}$, using $\|f\|^q_{L^q(\Omega_\rho)} = \rho^3\|\tilde f\|^q_{L^q(\Omega_1)}$. Similarly for $\nabla\mathcal{B}$. □

4.2. **The divergence-free test function.** Fix $\theta \in (0,1)$ throughout. For $\rho > 0$, let $\eta \in C_c^\infty(\mathbb{R}^3)$ be a radial cutoff function with

$$\eta \equiv 1 \text{ on } B_{\theta\rho}, \quad \operatorname{supp}\eta \subset B_\rho, \quad 0 \le \eta \le 1, \quad |\nabla\eta| \le \frac{C_\theta}{\rho}, \quad |\Delta\eta| \le \frac{C_\theta}{\rho^2}, \tag{7}$$

and $\operatorname{supp}\nabla\eta \subset A(\theta\rho,\rho)$. The constant C_θ depends only on θ. (Such an η is standard: take a smooth mollification of $\varphi(\cdot/\rho)^2$ supported in the annulus, or any radial bump satisfying the displayed bounds.)

Define the auxiliary field

$$b(x) := \mathcal{B}_{A(\theta\rho,\rho)}\big[v(\cdot)\cdot\nabla\eta(\cdot)\big](x), \qquad x \in \mathbb{R}^3, \tag{8}$$

extended by zero outside $A(\theta\rho,\rho)$. The argument $v\cdot\nabla\eta$ is integrable and has mean zero on $A(\theta\rho,\rho)$ because

$$\int_{A(\theta\rho,\rho)} v\cdot\nabla\eta\,dx = \int_{\mathbb{R}^3} v\cdot\nabla\eta\,dx = -\int_{\mathbb{R}^3}\eta\,(\operatorname{div} v)\,dx = 0,$$

so b is well defined by Lemma 4.1.

Lemma 4.5 (Test-function properties)**.** *The field $w := v\,\eta - b$ is divergence-free on $\mathbb{R}^3$, lies in $W^{1,q}(\mathbb{R}^3;\mathbb{R}^3)$ for every $q \in (1,\infty)$, and is compactly supported in $\overline{B_\rho}$.*

Proof. Support: $v\eta$ is supported in $\operatorname{supp}\eta \subset B_\rho$, and b is supported in $A(\theta\rho,\rho) \subset B_\rho$ by (B1).

Divergence:

$$\operatorname{div}(v\eta) = \eta\,\operatorname{div} v + v\cdot\nabla\eta = v\cdot\nabla\eta,$$

and by (B1), $\operatorname{div} b = v\cdot\nabla\eta$ on $A(\theta\rho,\rho)$; since both functions are zero outside $A(\theta\rho,\rho)$, we have $\operatorname{div} b = v\cdot\nabla\eta$ on all of $\mathbb{R}^3$ in the sense of distributions. Subtracting, $\operatorname{div} w = 0$.

Regularity: v and η are smooth, hence $v\eta \in C_c^\infty$, and $b \in W_0^{1,q}(A(\theta\rho,\rho);\mathbb{R}^3)$ by (B2) for every q. □

4.3. **The resulting identity.**

Proposition 4.6 (Pressure-eliminated identity)**.** *With η and w as above,*

$$\nu\int_{\mathbb{R}^3}|\nabla v|^2\,\eta\,dx = \frac{\nu}{2}\int_{\mathbb{R}^3}|v|^2\Delta\eta\,dx + \frac{1}{2}\int_{\mathbb{R}^3}|v|^2\,(v\cdot\nabla\eta)\,dx - \int_{\mathbb{R}^3}(v\otimes v)\colon\nabla b\,dx + \nu\int_{\mathbb{R}^3}\nabla v\colon\nabla b\,dx. \tag{9}$$

Proof. Pair the stationary Navier–Stokes equation $-\nu\Delta v + (v\cdot\nabla)v + \nabla p = 0$ with $w = v\eta - b$ and integrate over $\mathbb{R}^3$. Since $\operatorname{div} w = 0$ and w is compactly supported,

$$\int \nabla p\cdot w\,dx = -\int p\,\operatorname{div} w\,dx = 0.$$

So the identity reduces to

$$-\nu\int(\Delta v)\cdot w\,dx + \int(v\cdot\nabla v)\cdot w\,dx = 0. \tag{10}$$

Expand $w = v\eta - b$ and integrate by parts (valid by the compact support of w and smoothness of v):

$$-\int(\Delta v)_i\cdot(v_i\eta)\,dx = \int(\partial_j v_i)\,\partial_j(v_i\eta)\,dx = \int|\nabla v|^2\eta\,dx + \frac{1}{2}\int\nabla|v|^2\cdot\nabla\eta\,dx,$$

and a final integration by parts on the last term gives $-\frac{1}{2}\int|v|^2\Delta\eta\,dx$. Hence

$$-\nu\int(\Delta v)\cdot(v\eta)\,dx = \nu\int|\nabla v|^2\eta\,dx - \frac{\nu}{2}\int|v|^2\Delta\eta\,dx.$$

For the b-piece,

$$-\nu \int (\Delta v)\cdot(-b)\,dx = -\nu \int \nabla v\colon \nabla b\,dx,$$

by integration by parts (using $b \in W_0^{1,q}$ for every q). For the convective pairing against $v\eta$, use $\operatorname{div} v = 0$:

$$\int (v\cdot\nabla v)_i (v_i\eta)\,dx = \frac{1}{2}\int v\cdot\nabla|v|^2\,\eta\,dx = -\frac{1}{2}\int |v|^2\,(v\cdot\nabla\eta)\,dx,$$

the final IBP using $\operatorname{div} v = 0$ again. For the convective pairing against $-b$, use $\operatorname{div} v = 0$ once more:

$$-\int (v\cdot\nabla v)_i\, b_i\,dx = \int v_i v_j\,\partial_j b_i\,dx = \int (v\otimes v)\colon \nabla b\,dx.$$

Substituting the four pieces into (10) gives

$$\Big[\nu\int |\nabla v|^2\eta - \frac{\nu}{2}\int |v|^2\Delta\eta\Big] - \nu\int \nabla v\colon\nabla b + \Big[-\frac{1}{2}\int |v|^2 v\cdot\nabla\eta\Big] + \int (v\otimes v)\colon\nabla b = 0,$$

and rearrangement yields (9). □

Remark 4.7 (Why the Bogovskiĭ correction is forced)**.** A naive test function $v\eta$ is not divergence-free unless η is constant on $\operatorname{supp} v$; the correction $\mathcal{B}[v\cdot\nabla\eta]$ restores the constraint exactly. Without it, $\int \nabla p\cdot v\eta\,dx = -\int p\,v\cdot\nabla\eta\,dx$ carries a pressure remainder that cannot be bounded uniformly in ρ by any standard argument at the critical exponent. The trade is paid in the two new terms $\nu\int\nabla v\colon\nabla b$ and $\int (v\otimes v)\colon\nabla b$ on the right of (9); both are controlled in the next section.

5. Caccioppoli inequality on annuli

From the identity (9) we extract a Caccioppoli-type bound whose right-hand side is *sublinear* in $D(\rho)$. The restructuring avoids the Young-inequality absorption step that, in earlier drafts, forced a case dichotomy; it keeps cross terms as products $\|\nabla v\|_{L^2}\cdot$ (scale-corrected factor) bounded by direct Cauchy–Schwarz plus Lorentz embedding.

Throughout this section we take $\eta = \varphi(|\cdot|/\rho)$ as the hat function of Notation 2.1. The identity of Proposition 4.6 holds for this choice as the limit, as $\varepsilon\to 0$, of the smooth-mollification identity for $\eta_\varepsilon \in C_c^\infty$ satisfying (7); both sides of (9) converge by dominated convergence and the Bogovskiĭ linearity [10, III.3.1]. With this choice,

$$\int_{\mathbb{R}^3} |\nabla v|^2\eta\,dx = \mathcal{E}(\rho).$$

Proposition 5.1 (Sublinear Caccioppoli estimate)**.** *There is a constant $C_\star = C_\star(\theta,\nu,M)$ such that for every $\rho>0$,*

$$\nu\mathcal{E}(\rho) \le C_\star\rho^{-1/2}D(\rho)^{1/2} + \frac{\nu}{2}|I_1(\rho)| + \frac{1}{2}\int_{\mathbb{R}^3}|v|^2|v\cdot\nabla\eta|\,dx, \tag{11}$$

where the residual viscous term $|I_1(\rho)| \le C_\theta\nu M^2/\rho$, and the third term (the convective cutoff) is bounded by $C_\theta\rho^{-1}\|v\|^3_{L^3(A(\theta\rho,\rho))}$.

Proof. By (9) (rearranged):

$$\nu\mathcal{E}(\rho) \le \frac{\nu}{2}\Big|\int |v|^2\Delta\eta\,dx\Big| + \frac{1}{2}\Big|\int |v|^2 v\cdot\nabla\eta\,dx\Big| + \Big|\int (v\otimes v):\nabla b\,dx\Big| + \nu\Big|\int \nabla v:\nabla b\,dx\Big|.$$

We bound each of the four terms.

Term I_1 (viscous cutoff). Using $|\Delta\eta| \le C_\theta/\rho^2$ on $A(\theta\rho,\rho)$ (supp of $\Delta\eta$),

$$|I_1| := \Big|\int |v|^2\Delta\eta\,dx\Big| \le \frac{C_\theta}{\rho^2}\|v\|^2_{L^2(A(\theta\rho,\rho))}.$$

By the Lorentz layer-cake embedding proven in §6 (Proposition 6.1), $\|v\|^2_{L^2(A)} \le C\,M^2\,\rho$, so $|I_1| \le C_\theta M^2/\rho$, and $\frac{\nu}{2}|I_1| \le C_\theta \nu M^2/\rho$.

Term I_2 (Bogovskiĭ viscous). By Cauchy–Schwarz,

$$|I_2| := |\int \nabla v : \nabla b\,dx| \le \|\nabla v\|_{L^2(A)} \cdot \|\nabla b\|_{L^2(A)} = D(\rho)^{1/2} \cdot \|\nabla b\|_{L^2(A)}.$$

Using the uniform annular Bogovskiĭ bound Corollary 4.4 with $q = 2$ and $|\nabla\eta| \le C_\theta/\rho$,

$$\|\nabla b\|_{L^2(A)} \le C_\theta \|v \cdot \nabla\eta\|_{L^2(A)} \le \frac{C_\theta}{\rho}\|v\|_{L^2(A)} \le \frac{C_\theta M}{\sqrt{\rho}},$$

by the Lorentz bound of §6. Combining,

$$\nu|I_2| \le C_\theta \nu M\,\rho^{-1/2}\,D(\rho)^{1/2}.$$

Term I_4 (convective Bogovskiĭ). Apply the three-factor Hölder inequality with exponents $(12/5, 2, 12)$ to the identity $\int (v \otimes v) : \nabla b\,dx = \int v_i\,v_j\,\partial_j b_i\,dx$; equivalently (via the integration-by-parts identity $\int (v \otimes v) : \nabla b = -\int (v \cdot \nabla v) \cdot b$ from Proposition 4.6), bound

$$|I_4| := |\int (v \otimes v) : \nabla b\,dx| = |\int (v \cdot \nabla v) \cdot b\,dx| \le \|v\|_{L^{12/5}(A)}\,\|\nabla v\|_{L^2(A)}\,\|b\|_{L^{12}(A)}.$$

By the Sobolev embedding $W^{1,12/5}_0(A) \hookrightarrow L^{12}(A)$ (rescaled from the reference annulus $A_0 = A(\theta, 1)$, with constant depending only on θ),

$$\|b\|_{L^{12}(A)} \le C_\theta \|\nabla b\|_{L^{12/5}(A)}.$$

Bogovskiĭ at $q = 12/5$ (Corollary 4.4) gives

$$\|\nabla b\|_{L^{12/5}(A)} \le C_\theta \|v \cdot \nabla\eta\|_{L^{12/5}(A)} \le \frac{C_\theta}{\rho}\|v\|_{L^{12/5}(A)}.$$

By the Lorentz embedding (Proposition 6.1) at $r = 12/5$,

$$\|v\|_{L^{12/5}(A)} \le CM\rho^{3(5/12)-1} = CM\rho^{1/4}.$$

Assembling,

$$|I_4| \le (CM\rho^{1/4}) \cdot D(\rho)^{1/2} \cdot \Big(\frac{C_\theta}{\rho} \cdot CM\rho^{1/4}\Big) = C_\theta M^2 \rho^{-1/2} D(\rho)^{1/2}.$$

Setting $C_\star := C_\theta(\nu M + M^2)$ (so that $\nu|I_2| + |I_4| \le C_\star \rho^{-1/2} D^{1/2}$) and collecting the bounds on I_1, I_2, I_4, we obtain (11). The residual I_3-type term $\frac{1}{2}\int |v|^2 v \cdot \nabla\eta$ is bounded above by $\frac{C_\theta}{\rho}\|v\|^3_{L^3(A)}$ using $|\nabla\eta| \le C_\theta/\rho$ on the support A. □

The third term of (11) involves the scale-critical quantity $\|v\|^3_{L^3(A)}$, which is not bounded directly by Lorentz (the Lorentz embedding of §6 requires $r < 3$ strictly). This term is handled by the Gagliardo–Nirenberg interpolation of §7, which factors it through the sub-scale-critical Lorentz bounds plus the gradient norm.

Remark 5.2 (Draft-9 sublinearity vs. Young absorption)**.** Earlier approaches to stationary Liouville (Galdi's original argument, and Drafts 7–8 in the companion corpus) attempted to absorb the Bogovskiĭ-viscous cross-term into the left-hand side by Young's inequality: $ab \le \frac{1}{2}a^2 + \frac{1}{2}b^2$ with the a^2-piece absorbed into $D(\rho)$. This requires D to appear on the left of (11), which is not the case here: the left-hand side is $\mathcal{E}(\rho)$, not D. The identity (9) cannot be rearranged to put D on the left without introducing a Caccioppoli inequality for D itself, which is not available at this regularity. The sublinear form (11) with $D^{1/2}$ on the right is therefore the correct structural

form, confirmed by the explicit Bogovskiĭ scaling. The retired Young-absorption approach is catalogued as anti-knowledge AK-28 in the companion corpus.

6. Lorentz layer-cake embedding on annuli

Recall $\|f\|_{L^{q,\infty}(\mathbb{R}^n)} := \sup_{t>0} t\,|\{|f| > t\}|^{1/q}$. For $f \in L^{q,\infty}$ restricted to a set E of finite measure, interpolation with L^∞ gives an L^r bound for every $r \in (0, q)$.

Proposition 6.1 (Lorentz embedding on sets of finite measure)**.** *Let $f \in L^{q,\infty}(\mathbb{R}^n)$ with $q \in (0,\infty)$, and let $E \subset \mathbb{R}^n$ be a measurable set of finite measure. For every $r \in (0, q)$,*

$$\|f\|_{L^r(E)} \le \Big(\frac{q}{q-r}\Big)^{1/r} \|f\|_{L^{q,\infty}(\mathbb{R}^n)}\,|E|^{1/r-1/q}. \tag{12}$$

Proof. Write $M := \|f\|_{L^{q,\infty}}$. By Cavalieri's identity,

$$\int_E |f|^r\,dx = r\int_0^\infty t^{r-1}\,|\{x \in E : |f(x)| > t\}|\,dt.$$

Bound the distribution function by the minimum of its two available estimates:

$$|\{x \in E : |f(x)| > t\}| \le \min\Big(|E|,\, M^q/t^q\Big).$$

The two bounds are equal at $t = t_* := (M^q/|E|)^{1/q}$. Splitting the integral at t_*,

$$\int_E |f|^r\,dx \le r\,|E|\int_0^{t_*} t^{r-1}\,dt + r\,M^q\int_{t_*}^\infty t^{r-1-q}\,dt = |E|\,t_*^r + \frac{r}{q-r}\,M^q\,t_*^{r-q}.$$

Substituting $t_*^r = (M^q/|E|)^{r/q}$ and $t_*^{r-q} = (|E|/M^q)^{1-r/q}$,

$$\int_E |f|^r\,dx \le M^r|E|^{1-r/q}\Big(1 + \frac{r}{q-r}\Big) = \frac{q}{q-r}\,M^r\,|E|^{1-r/q}.$$

Taking the r-th root yields (12). □

We apply (12) with $q = 3$, $f = v$, and $E = A(\theta\rho, \rho)$. The volume of the annulus is

$$|A(\theta\rho,\rho)| = \tfrac{4\pi}{3}(1-\theta^3)\rho^3 =: c_\theta\rho^3.$$

Corollary 6.2 (Lorentz bound on the annulus)**.** *For every $r \in (0, 3)$ and every $\rho > 0$,*

$$\|v\|_{L^r(A(\theta\rho,\rho))} \le C_{r,\theta}\,M\,\rho^{3/r-1}. \tag{13}$$

Proof. Apply Proposition 6.1 with $q = 3$: $\|v\|_{L^r(A)} \le (3/(3-r))^{1/r} M\cdot(c_\theta\rho^3)^{1/r-1/3} = C_{r,\theta}M\rho^{3/r-1}$. □

Remark 6.3 (Why the embedding fails at $r = 3$)**.** As $r \to 3^-$, the prefactor $(3/(3-r))^{1/r}$ diverges, and the exponent $3/r - 1 \to 0$. The Lorentz quasi-norm $\|\cdot\|_{L^{3,\infty}}$ is not absolutely continuous: a function like $v(x) = |x|^{-1}$ has $\|v\|_{L^{3,\infty}(\mathbb{R}^3)} < \infty$ but $\|v\|_{L^3(B_R\setminus B_r)} = c\log^{1/3}(R/r) \to \infty$ as $r \to 0$. This is the structural reason Theorem 1.1 requires the sub-scale-critical machinery of §7: at $r = 3$, no Lorentz-based bound is available, and the critical exponent must be accessed through the gradient via Gagliardo–Nirenberg.

The key bounds used in §5 are the specific instances $r = 2$ (giving $\|v\|_{L^2(A)} \le CM\rho^{1/2}$) and $r = 12/5$ (giving $\|v\|_{L^{12/5}(A)} \le CM\rho^{1/4}$). The sub-scale-critical range $r \in (3/2, 3)$ underlies §7.

7. Gagliardo–Nirenberg interpolation and scale cancellation

The convective cutoff term in (11) is cubic in v and requires an $L^3(A)$ bound on v. The Lorentz embedding of §6 does not reach $r = 3$ endpoint. We therefore interpolate $\|v\|^3_{L^3(A)}$ between a sub-critical Lorentz bound and the annular gradient norm, via Gagliardo–Nirenberg, and observe that the exact scale-invariance of the critical L^3 exponent forces a cancellation of ρ-powers in the main term.

7.1. Reference Gagliardo–Nirenberg inequality.

Lemma 7.1 (Gagliardo–Nirenberg on the reference annulus)**.** *Fix $\theta \in (0,1)$ and $p \in (3/2, 3)$, and set $\theta_p := p/(6-p) \in (0,1)$. There is a constant $C_{\mathrm{GN}}(p,\theta)$ such that for every $w \in W^{1,2}(A(\theta,1)) \cap L^p(A(\theta,1))$,*

$$\|w\|_{L^3(A(\theta,1))} \le C_{\mathrm{GN}}\big(\|w\|^{\theta_p}_{L^p(A(\theta,1))}\|\nabla w\|^{1-\theta_p}_{L^2(A(\theta,1))} + \|w\|_{L^p(A(\theta,1))}\big). \tag{14}$$

Proof. This is the standard multiplicative Gagliardo–Nirenberg inequality on a bounded Lipschitz domain [10, Thm II.5.2]. The exponent θ_p is determined by the scaling relation

$$\frac{1}{3} = \frac{\theta_p}{p} + (1-\theta_p)\Big(\frac{1}{2} - \frac{1}{3}\Big),$$

which rearranges to $\theta_p = p/(6-p)$. The additive $\|w\|_{L^p}$ term is required on bounded domains because $\|w\|_{L^p} + \|\nabla w\|_{L^2}$ bounds the full $W^{1,p} \cap W^{1,2}$ norm only up to this lower-order piece. The constant C_{GN} depends on the annulus $A(\theta,1)$ through standard Sobolev-extension estimates. □

Cubing (14) and using $(x+y)^3 \le 4(x^3+y^3)$,

$$\|w\|^3_{L^3(A(\theta,1))} \le C'_{\mathrm{GN}}\Big(\|w\|^a_{L^p(A(\theta,1))}\,\|\nabla w\|^{2b}_{L^2(A(\theta,1))} + \|w\|^3_{L^p(A(\theta,1))}\Big), \tag{15}$$

where

$$a := 3\theta_p = \frac{3p}{6-p}, \qquad b := \frac{3(1-\theta_p)}{2} = \frac{3(3-p)}{6-p}, \qquad a + 2b = 3.$$

7.2. Rescaling to $A(\theta\rho,\rho)$ and scale cancellation.

Proposition 7.2 (Critical GN bound with exact ρ-cancellation)**.** *For every $\theta \in (0,1)$ and $p \in (3/2,3)$, and every $\rho > 0$,*

$$\|v\|^3_{L^3(A(\theta\rho,\rho))} \le C'_{\mathrm{GN}}\Big(\|v\|^a_{L^p(A(\theta\rho,\rho))}\,D(\rho)^b + \|v\|^3_{L^p(A(\theta\rho,\rho))}\Big), \tag{16}$$

with no explicit ρ-factor. Substituting the Lorentz bound Corollary 6.2 yields

$$\|v\|^3_{L^3(A(\theta\rho,\rho))} \le C'_{\mathrm{GN}}(CM)^a\,\rho^b\,D(\rho)^b + C'_{\mathrm{GN}}(CM)^3. \tag{17}$$

Proof. Let $\tilde{v}(y) := v(\rho y)$ for $y \in A(\theta,1)$. Direct substitution in each integral gives

$$\|\tilde{v}\|^q_{L^q(A(\theta,1))} = \rho^{-3}\,\|v\|^q_{L^q(A(\theta\rho,\rho))}, \qquad \|\nabla_y \tilde{v}\|^2_{L^2(A(\theta,1))} = \rho^{-1}\,D(\rho),$$

for any $q \ge 1$. Substituting into (15) applied to $\tilde{v}$,

$$\rho^{-3}\,\|v\|^3_{L^3(A)} \le C'_{\mathrm{GN}}\big(\rho^{-3a/p-b}\|v\|^a_{L^p(A)}\,D(\rho)^b + \rho^{-9/p}\|v\|^3_{L^p(A)}\big).$$

Multiplying by ρ^3, the first prefactor becomes $\rho^{3-3a/p-b}$. Substitute $a = 3p/(6-p)$, $b = 3(3-p)/(6-p)$:

$$3 - \frac{3a}{p} - b = 3 - \frac{9}{6-p} - \frac{3(3-p)}{6-p} = \frac{3(6-p) - 9 - 9 + 3p}{6-p} = \frac{18 - 3p - 18 + 3p}{6-p} = 0.$$

The second prefactor becomes

$$\rho^{3-9/p}\,\|v\|^3_{L^p(A)} \le \rho^{3-9/p}\cdot (CM)^3\rho^{3(3/p-1)} = (CM)^3\,\rho^{3-9/p+9/p-3} = (CM)^3,$$

using Lorentz $\|v\|_{L^p(A)} \le CM\rho^{3/p-1}$. This proves (16) and, with the Lorentz substitution $\|v\|^a_{L^p(A)} \le (CM)^a\rho^{a(3/p-1)}$ where $a(3/p-1)=b$ by direct computation, yields (17). □

7.3. **The full key estimate.**

Theorem 7.3 (Main sublinear estimate on $\mathcal{E}$)**.** *Fix $\theta\in(0,1)$ and $p\in(3/2,3)$, with $\gamma := (3-2p)/(6-p)$ and $b := 3(3-p)/(6-p)$. There are constants K_0,K_1,K_2 depending only on (θ,ν,M,p) such that, for every $\rho>0$,*

$$\mathcal{E}(\rho) \le K_0\,\rho^{-1/2}\,D(\rho)^{1/2} + K_1\,\rho^{-1} + K_2\,\rho^{\gamma}\,D(\rho)^b. \tag{18}$$

Proof. Divide (11) (Proposition 5.1) by ν:

$$\mathcal{E}(\rho) \le \frac{C_\star}{\nu}\,\rho^{-1/2}D^{1/2} + \frac{1}{2}|I_1(\rho)| + \frac{1}{2\nu}\int |v|^2|v\cdot\nabla\eta|\,dx.$$

The first term contributes $K_0\rho^{-1/2}D^{1/2}$ with $K_0 = C_\star/\nu$.

The second term is bounded by $|I_1| \le C_\theta M^2/\rho$, contributing $C_\theta M^2/(2\rho)$.

The third term, using $|\nabla\eta| \le C_\theta/\rho$ on $A(\theta\rho,\rho)$,

$$\int |v|^2|v\cdot\nabla\eta|\,dx \le \frac{C_\theta}{\rho}\,\|v\|^3_{L^3(A(\theta\rho,\rho))}.$$

Applying (17),

$$\frac{1}{\rho}\,\|v\|^3_{L^3(A)} \le C'_{\mathrm{GN}}(CM)^a\,\rho^{b-1}\,D^b + \frac{C'_{\mathrm{GN}}(CM)^3}{\rho}.$$

Since $b-1=\gamma$,

$$\frac{1}{2\nu}\int |v|^2|v\cdot\nabla\eta|\,dx \le \frac{C_\theta C'_{\mathrm{GN}}C^aM^a}{2\nu}\,\rho^{\gamma}\,D^b + \frac{C_\theta C'_{\mathrm{GN}}C^3M^3}{2\nu\rho}.$$

Combining the ρ^{-1} contributions into K_1, the $\rho^\gamma D^b$ contribution into K_2, and the $\rho^{-1/2}D^{1/2}$ contribution into K_0 gives (18). □

Remark 7.4 (Exponents at $p=2$)**.** The natural choice $p=2$ gives

$$a = \frac{6}{4} = \frac{3}{2},\quad b = \frac{3}{4},\quad \gamma = -\frac{1}{4}.$$

The exponents $(\gamma,b) = (-1/4,3/4)$ are the ones used in the Draft-9 closure of the Caccioppoli restructuring in the companion corpus; they give the cleanest integration in §8.

8. The superlinear differential inequality

We reduce Theorem 7.3 to a closed ODE for $\mathcal{E}$. Throughout this section we fix $p\in(3/2,12/5]$ so that the critical GN exponent $b = 3(3-p)/(6-p) \ge 1/2$, and we will focus on the canonical choice $p=2$ for the explicit exponents.

Proposition 8.1 (Superlinear ODE)**.** *Fix $p\in(3/2,12/5]$, with $\gamma = (3-2p)/(6-p)$ and $b = 3(3-p)/(6-p)$. If $\lim_{\rho\to\infty}\mathcal{E}(\rho) = +\infty$, then there exist $c>0$ and $\rho_*\ge 1$, depending on (θ,ν,M,p), such that*

$$\mathcal{E}'(\rho) \ge c\,\rho^{-\alpha}\,\mathcal{E}(\rho)^\beta \qquad \text{for all } \rho\ge\rho_*, \tag{19}$$

with

$$\alpha = 1 + \gamma/b = \frac{12-5p}{3(3-p)}, \qquad \beta = \frac{1}{b} = \frac{6-p}{3(3-p)}.$$

Moreover, $\alpha < 1$ *and* $\beta > 1$ *for every* $p \in (3/2, 3)$.

Proof. Since $b \geq 1/2$, for any $D \geq 0$,

$$D^{1/2} \leq D^b + 1,$$

distinguishing $D \geq 1$ (where $D^{1/2} \leq D^b$) from $D < 1$ (where $D^{1/2} \leq 1$). Substituting into (18):

$$\mathcal{E}(\rho) \leq K_0\rho^{-1/2}(D(\rho)^b + 1) + K_1\rho^{-1} + K_2\rho^\gamma D(\rho)^b = (K_0\rho^{-1/2} + K_2\rho^\gamma)D(\rho)^b + R(\rho), \tag{20}$$

where $R(\rho) := K_0\rho^{-1/2} + K_1\rho^{-1}$. Since $\gamma \geq -1/2$ for $p \in (3/2, 12/5]$ (explicit check: $\gamma = -1/2$ at $p = 12/5$; γ decreases with p from 0 at $p = 3/2$), for $\rho \geq 1$ we have $\rho^{-1/2} \leq \rho^\gamma$, so

$$\mathcal{E}(\rho) \leq (K_0 + K_2)\rho^\gamma D(\rho)^b + R(\rho).$$

Set $K := K_0 + K_2$.

By the hypothesis $\mathcal{E} \to \infty$, there is $\rho_1 \geq 1$ such that $\mathcal{E}(\rho) \geq 2R(\rho)$ for all $\rho \geq \rho_1$ (since $R(\rho) \to 0$ and $\mathcal{E}(\rho) \to \infty$). Then

$$\mathcal{E}(\rho) - R(\rho) \geq \tfrac{1}{2}\mathcal{E}(\rho), \quad \text{so} \quad \tfrac{1}{2}\mathcal{E}(\rho) \leq K\rho^\gamma D(\rho)^b \quad \forall \rho \geq \rho_1.$$

Rearranging,

$$D(\rho) \geq \Big(\frac{\mathcal{E}(\rho)}{2K}\Big)^{1/b} \rho^{-\gamma/b}.$$

By Lemma 2.2, $D(\rho) \leq \frac{1-\theta}{\theta}\,\rho\,\mathcal{E}'(\rho)$, so

$$\rho\,\mathcal{E}'(\rho) \geq \frac{\theta}{1-\theta}\Big(\frac{\mathcal{E}(\rho)}{2K}\Big)^{1/b} \rho^{-\gamma/b}.$$

Divide by ρ:

$$\mathcal{E}'(\rho) \geq c\,\rho^{-1-\gamma/b}\,\mathcal{E}(\rho)^{1/b}, \qquad c := \frac{\theta}{(1-\theta)(2K)^{1/b}}.$$

This is (19) with $\alpha = 1 + \gamma/b$ and $\beta = 1/b$, valid for $\rho \geq \rho_* := \rho_1$.

It remains to verify $\alpha < 1$ and $\beta > 1$. Substituting $\gamma = (3-2p)/(6-p)$ and $b = 3(3-p)/(6-p)$,

$$\frac{\gamma}{b} = \frac{(3-2p)/(6-p)}{3(3-p)/(6-p)} = \frac{3-2p}{3(3-p)}, \qquad \alpha = 1 + \frac{3-2p}{3(3-p)} = \frac{3(3-p)+3-2p}{3(3-p)} = \frac{12-5p}{3(3-p)},$$

$$\beta = \frac{6-p}{3(3-p)}.$$

For $p \in (3/2, 3)$, the denominator $3(3-p) > 0$. Then

$$\alpha < 1 \iff 12 - 5p < 3(3-p) = 9 - 3p \iff 2p > 3 \iff p > 3/2,$$

$$\beta > 1 \iff 6 - p > 3(3-p) = 9 - 3p \iff 2p > 3 \iff p > 3/2.$$

Both conditions hold exactly when $p > 3/2$. □

Corollary 8.2 (Explicit exponents at $p = 2$)**.** *At* $p = 2$,

$$\gamma = -\tfrac{1}{4}, \quad b = \tfrac{3}{4}, \quad \alpha = \tfrac{2}{3}, \quad \beta = \tfrac{4}{3}.$$

Remark 8.3 (Sharp failure at $p = 3/2$ and $p = 3$)**.** The endpoint $p = 3/2$ corresponds to $\alpha = \beta = 1$, where the superlinear structure degenerates (the ODE becomes linear and §9 does not close). The endpoint $p = 3$ lies outside the Lorentz embedding range of §6. The open range $p \in (3/2, 3)$ is therefore the full admissible set; the argument is strongest at p away from both endpoints.

9. Integration and contradiction

We integrate the ODE of Proposition 8.1 and derive a contradiction with the non-decreasing, finite-at-each-scale behaviour of $\mathcal{E}$.

Theorem 9.1 (Finite Dirichlet energy)**.** $\int_{\mathbb{R}^3} |\nabla v|^2\, dx < \infty$.

Proof. By Lemma 2.2, $\mathcal{E}$ is non-decreasing, so either $\lim_{\rho\to\infty} \mathcal{E}(\rho) = L < \infty$ or $\lim_{\rho\to\infty} \mathcal{E}(\rho) = +\infty$. In the first case,

$$\int_{\mathbb{R}^3} |\nabla v|^2\, dx = \lim_{\rho\to\infty} \int |\nabla v|^2\, \varphi(|x|/\rho)\, dx = L < \infty,$$

by monotone convergence ($\varphi(|x|/\rho) \nearrow 1$ pointwise as $\rho \to \infty$).

It remains to rule out the second case. Assume, for contradiction, $\mathcal{E}(\rho) \to +\infty$. Proposition 8.1 applies: for some $c > 0$, $\rho_* \geq 1$, and (α, β) with $\alpha < 1 < \beta$ (at $p = 2$: $\alpha = 2/3$, $\beta = 4/3$),

$$\mathcal{E}'(\rho) \geq c\, \rho^{-\alpha}\, \mathcal{E}(\rho)^{\beta} \qquad \forall \rho \geq \rho_*.$$

Since $\mathcal{E}(\rho) > 0$ (on the hypothesis $\mathcal{E} \to \infty$ we may assume $\mathcal{E}(\rho_*) > 0$), divide by $\mathcal{E}^{\beta}$:

$$\mathcal{E}(\rho)^{-\beta}\, \mathcal{E}'(\rho) \geq c\, \rho^{-\alpha} \qquad \forall \rho \geq \rho_*.$$

Integrate from ρ_* to $R > \rho_*$:

$$\int_{\rho_*}^{R} \mathcal{E}(\rho)^{-\beta}\, \mathcal{E}'(\rho)\, d\rho \geq c \int_{\rho_*}^{R} \rho^{-\alpha}\, d\rho.$$

The left-hand side computes (using $\beta > 1$, so $1 - \beta < 0$):

$$\int_{\rho_*}^{R} \mathcal{E}^{-\beta}\, \mathcal{E}'\, d\rho = \Big[\frac{\mathcal{E}^{1-\beta}}{1-\beta}\Big]_{\rho_*}^{R} = \frac{1}{\beta - 1}\big(\mathcal{E}(\rho_*)^{1-\beta} - \mathcal{E}(R)^{1-\beta}\big) \leq \frac{\mathcal{E}(\rho_*)^{1-\beta}}{\beta - 1},$$

bounded uniformly in R because $\mathcal{E}(R)^{1-\beta} > 0$.

The right-hand side, using $\alpha < 1$:

$$c \int_{\rho_*}^{R} \rho^{-\alpha}\, d\rho = \frac{c}{1-\alpha}\big(R^{1-\alpha} - \rho_*^{1-\alpha}\big) \xrightarrow{R\to\infty} +\infty.$$

Comparing: a bounded quantity exceeds an unbounded one, which is impossible. The contradiction rules out $\mathcal{E}(\rho) \to \infty$. Therefore $\mathcal{E}$ is bounded and $\int_{\mathbb{R}^3} |\nabla v|^2 dx < \infty$. □

Remark 9.2 (Quantitative form of the bound)**.** The proof of Theorem 9.1 yields an explicit quantitative bound on $\mathcal{E}$: from $\frac{1}{\beta-1}\mathcal{E}(\rho_*)^{1-\beta} \geq \frac{c}{1-\alpha}(R^{1-\alpha} - \rho_*^{1-\alpha})$ taken in the contrapositive direction, any large value of $\mathcal{E}(\rho_*)$ is constrained by the divergence rate of $R^{1-\alpha}$. This gives the scale estimate $\mathcal{E}(\rho_0) \leq O(M^6/\rho_0)$ exploited in Corollary 1.4; see §11.

10. From finite Dirichlet energy to $v \equiv 0$

We close the proof of Theorem 1.1. The finite Dirichlet energy from Theorem 9.1, combined with $v \in L^{3,\infty}(\mathbb{R}^3)$, upgrades v to global L^6 regularity and the pressure p to global L^3; a global cutoff test then forces $\nabla v \equiv 0$, hence $v \equiv 0$.

Lemma 10.1 (Upgraded regularity)**.** *From $v \in L^{3,\infty}(\mathbb{R}^3)(\mathbb{R}^3)$ and $\int_{\mathbb{R}^3} |\nabla v|^2 dx < \infty$, it follows that $v \in L^6(\mathbb{R}^3)$ with $\|v\|_{L^6} \leq C\|\nabla v\|_{L^2}$, and the pressure p associated with the stationary NS system satisfies $p \in L^3(\mathbb{R}^3)$ with $\|p\|_{L^3} \leq C\|\nabla v\|_{L^2}^2$.*

Proof. The homogeneous Sobolev inequality on $\mathbb{R}^3$ [10, II.5] gives

$$\|v\|_{L^6(\mathbb{R}^3)} \le C_{\mathrm{Sob}}\,\|\nabla v\|_{L^2(\mathbb{R}^3)}.$$

(Strictly: v is assumed smooth, so Sobolev applies pointwise to $v - v_R$ on any B_R with v_R the mean, and passing $R \to \infty$ uses $v \in L^{3,\infty}(\mathbb{R}^3)$ to eliminate the constant.) Hence $v \otimes v \in L^3(\mathbb{R}^3)$ with $\|v \otimes v\|_{L^3} = \|v\|_{L^6}^2 \le C\|\nabla v\|_{L^2}^2$.

The stationary NS equation $-\nu\Delta v + \operatorname{div}(v \otimes v) + \nabla p = 0$ gives, taking divergence and using $\operatorname{div} v = 0$,

$$-\Delta p = \operatorname{div}\operatorname{div}(v \otimes v) = \partial_i\partial_j(v_i v_j).$$

In terms of Riesz transforms $R_i = (-\Delta)^{-1/2}\partial_i$, $p = R_i R_j(v_i v_j)$ modulo the additive constant, which is fixed to zero by requiring $p \in L^3$ (no nonzero constant lies in $L^3(\mathbb{R}^3)$). Calderón–Zygmund boundedness of $R_i R_j$ on $L^3(\mathbb{R}^3)$ gives $\|p\|_{L^3} \le C\|v \otimes v\|_{L^3} \le C\|\nabla v\|_{L^2}^2$. □

Proof of Theorem 1.1. Let $\varphi_R \in C_c^\infty(\mathbb{R}^3)$ be a radial cutoff with $\varphi_R \equiv 1$ on B_R, $\operatorname{supp}\varphi_R \subset B_{2R}$, $|\nabla\varphi_R| \le C/R$, and $\operatorname{supp}\nabla\varphi_R \subset A(R, 2R)$. Test the stationary NS equation against $v\,\varphi_R^2$ and integrate over $\mathbb{R}^3$:

$$\int \Big[-\nu\,\Delta v + (v \cdot \nabla)v + \nabla p\Big] \cdot v\,\varphi_R^2\,dx = 0.$$

Each term integrates by parts (valid by v smooth, φ_R compactly supported):

$$-\nu \int (\Delta v) \cdot v\,\varphi_R^2 = \nu \int |\nabla v|^2\varphi_R^2 + \nu \int \varphi_R\,(\nabla|v|^2) \cdot \nabla\varphi_R,$$

$$\int (v \cdot \nabla v) \cdot v\,\varphi_R^2 = -\int |v|^2\,(v \cdot \nabla\varphi_R)\,\varphi_R,$$

$$\int \nabla p \cdot v\,\varphi_R^2 = -2\int p\,(v \cdot \nabla\varphi_R)\,\varphi_R,$$

the last two using $\operatorname{div} v = 0$. Rearranging gives the identity

(21)
$$\nu \int |\nabla v|^2\varphi_R^2\,dx = -\nu \int \varphi_R\,(\nabla|v|^2) \cdot \nabla\varphi_R\,dx + \int |v|^2\,(v \cdot \nabla\varphi_R)\,\varphi_R\,dx + 2\int p\,(v \cdot \nabla\varphi_R)\,\varphi_R\,dx.$$

Denote the three right-hand terms J_1, J_2, J_3. Each is supported in $A(R, 2R)$.

Bound on J_1 (viscous cross-term). By Cauchy–Schwarz and $|\nabla\varphi_R| \le C/R$,

$$|J_1| = 2\nu\Big|\int \varphi_R\,(v \cdot \nabla v) \cdot \nabla\varphi_R\,dx\Big| \le \frac{2\nu C}{R}\,\|v\|_{L^2(A(R,2R))}\,\|\nabla v\|_{L^2(A(R,2R))}.$$

By Corollary 6.2 at $r = 2$, $\|v\|_{L^2(A(R,2R))} \le CMR^{1/2}$. Since $\int_{\mathbb{R}^3} |\nabla v|^2 < \infty$, $\|\nabla v\|_{L^2(A(R,2R))}^2 \le \int_{|x|\ge R} |\nabla v|^2\,dx \to 0$ as $R \to \infty$. Hence

$$|J_1| \le \frac{C\nu M}{R^{1/2}}\,\|\nabla v\|_{L^2(A(R,2R))} \xrightarrow{R\to\infty} 0.$$

Bound on J_2 (convective). Using $|\nabla\varphi_R| \le C/R$,

$$|J_2| \le \frac{C}{R}\int_{A(R,2R)} |v|^3\,dx = \frac{C}{R}\,\|v\|_{L^3(A(R,2R))}^3.$$

By Proposition 7.2 applied on $A(R, 2R)$ (i.e., with $\theta = 1/2$ and $\rho = 2R$),

$$\|v\|_{L^3(A)}^3 \le C'_{\mathrm{GN}}\big(M^a\,(2R)^b\,D_R^b + M^3\big), \qquad D_R := \|\nabla v\|_{L^2(A(R,2R))}^2,$$

with $a = 3/2$, $b = 3/4$ at $p = 2$. So

$$|J_2| \le \frac{C}{R}\big(CM^{3/2}(2R)^{3/4}D_R^{3/4} + CM^3\big) = CM^{3/2}R^{-1/4}\,D_R^{3/4} + CM^3R^{-1} \xrightarrow{R\to\infty} 0,$$

using $D_R \to 0$ and $R^{-1} \to 0$.

Bound on J_3 (pressure). Using $|\nabla\varphi_R| \le C/R$ and Hölder with $(3, 3/2)$,

$$|J_3| \le \frac{2C}{R}\int_{A(R,2R)} |p|\,|v|\,dx \le \frac{2C}{R}\,\|p\|_{L^3(A(R,2R))}\,\|v\|_{L^{3/2}(A(R,2R))}.$$

By Corollary 6.2 at $r = 3/2$, $\|v\|_{L^{3/2}(A(R,2R))} \le CMR$. And by Lemma 10.1, $p \in L^3(\mathbb{R}^3)$, so $\|p\|_{L^3(A(R,2R))} \le \|p\|_{L^3(|x|\ge R)} \to 0$ as $R \to \infty$ by absolute continuity of the L^3 integral. Hence

$$|J_3| \le \frac{2C\cdot CMR}{R}\,\|p\|_{L^3(|x|\ge R)} = 2C^2M\,\|p\|_{L^3(|x|\ge R)} \xrightarrow{R\to\infty} 0.$$

Passage to the limit. From (21),

$$\nu\int |\nabla v|^2\varphi_R^2\,dx = J_1 + J_2 + J_3,$$

and the left-hand side converges to $\nu\int_{\mathbb{R}^3}|\nabla v|^2dx$ as $R \to \infty$ by monotone convergence (since $\varphi_R^2 \to 1$ pointwise, $0 \le \varphi_R^2 \le 1$, $|\nabla v|^2 \in L^1$). The right-hand side converges to 0. Thus

$$\nu\int_{\mathbb{R}^3}|\nabla v|^2\,dx = 0, \qquad \text{hence } \nabla v \equiv 0, \qquad \text{hence } v \equiv \text{const}.$$

A nonzero constant does not lie in $L^{3,\infty}(\mathbb{R}^3)(\mathbb{R}^3)$: the distribution function $|\{x \in \mathbb{R}^3 : |v(x)| > t\}|$ is ∞ for $0 < t < |v|$ and 0 for $t > |v|$, so $\sup_t t\,|\{|v| > t\}|^{1/3} = \infty$. By hypothesis $\|v\|_{L^{3,\infty}(\mathbb{R}^3)} < \infty$, so the constant is zero: $v \equiv 0$. □

11. Extensions

11.1. **Four-dimensional Liouville.** The proof of Theorem 1.1 transposes to $\mathbb{R}^n$ with $n \ge 3$ by tracking the dimensional discriminant. The Gagliardo–Nirenberg scale-invariance computation of §7 gives, in dimension n at critical exponent L^n,

$$b = b(n,p) = \frac{n(n-p)}{2n - p(n-2)}, \qquad a(n,p) = n - 2b.$$

Substituting into the ODE exponents,

$$\gamma(n,p) = -\frac{(n-4)(n-2p)}{2n-p(n-2)}, \qquad \alpha(n,p) = 1 + \frac{\gamma(n,p)}{b(n,p)}, \qquad \beta(n,p) = \frac{1}{b(n,p)}.$$

At $n = 3$ these reduce to $\gamma(3,p) = (3-2p)/(6-p)$ and $b(3,p) = 3(3-p)/(6-p)$, recovering the exponents of Theorem 7.3 and Proposition 8.1. The factor $(n-4)$ in the numerator of $\gamma(n,p)$ governs the transition across dimension four.

Proof of Corollary 1.2. At $n = 4$, $\gamma(4,p) = 0$ for every p; the GN main term is scale-invariant and no ρ-weight survives. Taking the natural choice $p = 5/2$ (symmetric about the critical $L^{4,\infty}$): $b = 4\cdot(3/2)/(8 - 5/2\cdot 2) = 4\cdot(3/2)/3 = 2$; $\alpha = 1$; $\beta = 1/2$. This gives a borderline ODE $\mathcal{E}'(\rho) \ge c\,\rho^{-1}\,\mathcal{E}(\rho)^\beta$ with $\alpha = 1$, which is integrable on the left (bounded antiderivative since $\beta > 1$ — for p slightly larger than $5/2$, say $p \in (5/2, 4)$, one obtains $\beta > 1$ by direct substitution) and logarithmically divergent on the right:

$$c\int_{\rho_*}^{R}\frac{d\rho}{\rho} = c\log\frac{R}{\rho_*} \xrightarrow{R\to\infty} +\infty.$$

The contradiction closes the same way as in Theorem 9.1, forcing $\int_{\mathbb{R}^4}|\nabla v|^2dx < \infty$. The closing argument of §10 extends verbatim with Sobolev $\|v\|_{L^4(\mathbb{R}^4)} \le C\|\nabla v\|_{L^2(\mathbb{R}^4)}$ (critical Sobolev in dimension four) replacing the three-dimensional L^6 bound. Hence $v \equiv 0$ on $\mathbb{R}^4$. □

Remark 11.1 (Failure for $n \geq 5$)**.** For $n \geq 5$, $\gamma(n,p) > 0$ for p in the admissible range, making $\alpha(n,p) > 1$. The integral $\int \rho^{-\alpha} d\rho$ converges, and the RHS of the ODE integration stays bounded — no contradiction. A sharper structural obstruction applies: $\beta > 1$ requires $p > n/2$ (from the Sobolev embedding direction), while $\alpha \leq 1$ requires $p \leq n/2$ (from the integration direction); the two conditions are incompatible for $n \geq 5$. No stationary $L^{n,\infty}(\mathbb{R}^n)$ Liouville theorem is currently available for $n \geq 5$; the structural obstruction (Sobolev / integration exponent conflict) is discussed in the companion corpus [7].

11.2. The $W^{1,2}_{\text{loc}}$ weak-solution extension.

Proof of Corollary 1.3. Let $v \in L^{3,\infty}(\mathbb{R}^3)(\mathbb{R}^3) \cap W^{1,2}_{\text{loc}}(\mathbb{R}^3)$ be a weak solution of stationary Navier–Stokes in the sense that, for every divergence-free $\phi \in C_c^\infty(\mathbb{R}^3)$,

$$\nu \int \nabla v : \nabla \phi \, dx + \int v \otimes v : \nabla \phi \, dx = 0.$$

(The pressure is eliminated by the divergence-free test class.)

Step 1: local regularity. By $W^{1,2}_{\text{loc}}$ and the Sobolev embedding $W^{1,2}(B_R) \hookrightarrow L^6(B_R)$, $v \in L^6_{\text{loc}}$. Interpolating $v \in L^{3,\infty}(\mathbb{R}^3) \cap L^6_{\text{loc}}$ with Hölder gives $v \in L^r_{\text{loc}}$ for every $r \in [3,6]$. In particular $v \otimes v \in L^3_{\text{loc}}$, so $p \in L^3_{\text{loc}}$ by the local Calderón–Zygmund argument of Lemma 10.1 restricted to any B_R. Iteration with the equation $-\nu\Delta v + \operatorname{div}(v \otimes v) + \nabla p = 0$ (in the sense of distributions) gives $v \in C^\infty_{\text{loc}}$ (Galdi [10, Ch. V]).

Step 2: the Caccioppoli and GN machinery still apply. The test-function construction of §4 uses $w = v\eta - \mathcal{B}[v \cdot \nabla \eta]$; this requires only $v \in W^{1,q}_{\text{loc}}$ for some $q > 1$, which follows from the interior regularity of Step 1. Proposition 4.6 holds by mollification of v followed by passage to the limit, since v is smooth in the interior. The Lorentz and GN machinery of §6–§7 use $v \in L^{3,\infty}(\mathbb{R}^3)$ and local Sobolev regularity, both of which persist under the weak hypothesis.

Step 3: test-function density. The divergence-free test space $\{\phi \in C_c^\infty(\mathbb{R}^3) : \operatorname{div} \phi = 0\}$ is dense in the space $W^{1,q}_0$-divergence-free functions with compact support, for every $q > 1$ [10, Thm III.4.1] (originally due to Heywood [11]). The Bogovskiĭ-modified test function $w = v\eta - \mathcal{B}[v \cdot \nabla \eta]$ lies in this space by construction, so it is an admissible test function in the weak formulation.

Step 4: closing argument. The proof of Theorem 1.1 runs verbatim, concluding $v \equiv 0$. The $W^{1,2}_{\text{loc}}$ hypothesis is sharp: the Landau solutions [15] are smooth outside the origin, belong to $L^{3,\infty}(\mathbb{R}^3)(\mathbb{R}^3)$ (with quasi-norm equal to a definite constant), and lie in $W^{1,2}_{\text{loc}}(\mathbb{R}^3 \setminus \{0\})$ but not at the origin; they show that *any weakening below $W^{1,2}_{\text{loc}}$ admits nontrivial stationary solutions* in $L^{3,\infty}(\mathbb{R}^3)$. □

11.3. Quantitative bound.

Proof of Corollary 1.4. From the proof of Proposition 8.1, the ODE constant at $p = 2$ is

$$c = \frac{\theta}{(1-\theta)(2K_2)^{1/b}} = \frac{\theta}{(1-\theta)(2K_2)^{4/3}},$$

with $K_2 = O(M^{3/2})$ (tracking M-dependence through Theorem 7.3; the dominant M-power arises from $\|v\|^a_{L^p}$ with $a = 3/2$ at $p = 2$). So $c \geq c_0 M^{-2}$ for a dimensionless $c_0 = c_0(\theta, \nu)$.

From the integration of Theorem 9.1,

$$\frac{\mathcal{E}(\rho_*)^{1-\beta}}{\beta - 1} \geq \frac{c}{1-\alpha}\big(R^{1-\alpha} - \rho_*^{1-\alpha}\big) \qquad \forall R > \rho_*.$$

At $p = 2$: $\beta - 1 = 1/3$, $1 - \alpha = 1/3$. Taking $R = 2\rho_*$,

$$\mathcal{E}(\rho_*)^{-1/3} \geq \frac{c \cdot (2^{1/3} - 1)}{1/3 \cdot 1/3}\, \rho_*^{1/3} = C_0\, c\, \rho_*^{1/3}.$$

Inverting,

$$\mathcal{E}(\rho_*) \leq \frac{C_1}{c^3\, \rho_*} = \frac{C_1 M^6}{c_0^3\, \rho_*},$$

which is the claimed single-exponential-in-M bound.

Comparison with Tao. Tao's quantitative regularity bound [25], which targets the full Clay problem via an assumed uniform L^3-bound on the evolution, gives a triple-exponential dependence on the bound. Corollary 1.4 is, in principle, not directly comparable (it concerns the stationary case at a single scale), but the constructivized KNSS reduction using Theorem 1.1 as the Liouville step in place of a conjectured one would inherit the present polynomial-in-M bound. □

12. Application to the Clay chain and the $L^{3,\infty}$ endpoint gap

This section describes how Theorem 1.1 composes within the Koch–Nadirashvili–Seregin–Šverák blow-up framework, identifies the single structural step that remains open, and surveys the current partial closures.

12.1. **The composition.** Suppose $u \in C^\infty(\mathbb{R}^3 \times [0, T^*))$ is the maximal smooth solution of (1) with Schwartz u_0, and $T^* < \infty$. The chain runs:

- (C1) **Type I excluded.** NRS [18], Tsai [26], ESS [9]: no self-similar solutions; any hypothetical singularity is Type II.
- (C2) **KNSS rescaling.** [14, §6]: sequences $\lambda_n \to 0$, $x_n \to x_0$, $t_n \to T^*$ with $v_n(y, s) := \lambda_n u(x_n + \lambda_n y, t_n + \lambda_n^2 s) \to v$ in C^m_{loc}, $|v| \leq C_0$, $|v(0, 0)| = 1$.
- (C3) **Stationary backward limit.** Arzelà–Ascoli extracts $v(\cdot, s_n) \to v_\infty$ in C^m_{loc} along some $s_n \to -\infty$; equicontinuity of $\partial_s v$ forces v_∞ stationary.
- (C4) **Liouville applies.** Under the inheritance step discussed in §12.2, $v_\infty \in L^{3,\infty}(\mathbb{R}^3)$; by Theorem 1.1, $v_\infty \equiv 0$.
- (C5) **Backward uniqueness contradicts.** ESS [9, §5] + spatial continuation [9, §4] yield $\omega(\cdot, s) \neq 0$ for all $s \leq 0$, contradicting the full convergence from (C4).

Every step of this chain except (C4) is rigorous in the published literature. The exception is the implicit hypothesis that $v \in L^{3,\infty}(\mathbb{R}^3)$ uniformly in $s \leq 0$; the conclusion of Theorem 1.1 is conditional on that uniform bound.

12.2. **The inheritance gap.** Scale invariance of the $L^{3,\infty}(\mathbb{R}^3)$ quasi-norm shows that the ancient solution v inherits $\|v(\cdot, s)\|_{L^{3,\infty}(\mathbb{R}^3)}$ from $\|u(\cdot, t_n + \lambda_n^2 s)\|_{L^{3,\infty}(\mathbb{R}^3)}$. What remains is to bound the latter uniformly in $t \in [0, T^*)$. This is the question of the $L^{3,\infty}$ endpoint regularity criterion, which Phuc [22, p. 4] states as open:

> "To the best of our knowledge, a criterion of local regularity for the Navier–Stokes equations in $L^\infty(-1, 0; L^{3,\infty}(B_1))$ is still unknown."

The obstruction is structural. Phuc's method closes the criterion for $L^\infty_t L^{3,q}_x$ with $q \in [3, \infty)$ using exterior-tail compactness at the blow-up extraction: the rescaled ancient solution inherits $\|\cdot\|_{L^{3,q}(\mathbb{R}^3 \setminus B_R)} \to 0$ as $R \to \infty$ by absolute continuity of the $L^{3,q}$ norm. At $q = \infty$, the $L^{3,\infty}(\mathbb{R}^3)$ quasi-norm is not absolutely continuous — the function $|x|^{-1}$, the canonical critical profile, has scale-invariant exterior norm — and the compactness step collapses.

12.3. **Partial closures.**
Axisymmetric case. Ożański–Palasek [20] closed the endpoint for axisymmetric flows via Nazarov–Ural'tseva parabolic Harnack [19] applied to the swirl rv_θ, which satisfies a scalar parabolic equation with divergence-free drift. The reduction to a scalar quantity is essentially axisymmetric; no analog is known for general flows.
Smallness. Luo–Tsai [17] and Kim–Kozono [13] close the endpoint under a Serrin-type smallness assumption $\|u\|_{L_t^\infty L_x^{3,\infty}}$ below a threshold. The large-data question remains open.
Partial regularity. Seregin [24] established that an $L_t^\infty L_x^{3,\infty}$ bound controls the singular set to be finite (bounded by M^3/ϵ^4 for an effective ϵ); this is partial, not full, regularity.
Logarithmic improvement. Phuc [22, Thm 1.5] closes the endpoint under the pointwise refinement $|u(x,t)| \le C|x|^{-1}|\log(|x|/2)|^{-\beta}$ for any $\beta > 0$. A strengthening of the rescaled ancient-solution bound to this logarithmic form would complete the chain via Phuc's machinery.

12.4. **A weighted-moment proposal.** The obstruction at the L^2-only level is sharp: the translation-invariance of $\|\cdot\|_{L^2(\mathbb{R}^3)}$ forbids any constant depending only on $\|u_0\|_{L^2}$ from bounding $|u(x,t)|$ pointwise. Brandolese–Meyer [2] exhibit explicit counterexamples. The constant must depend on a weighted norm carrying spatial information.

A natural proposal is the weighted hypothesis $(1+|x|)u_0 \in L^\infty(\mathbb{R}^3)$, which is automatic for Schwartz data. Define $\mathcal{M}(u_0) := \|u_0\|_{L^2} + \|(1+|x|)u_0\|_{L^\infty}$. We conjecture that the rescaled ancient solution inherits $\|v(\cdot,s)\|_{L^{3,\infty}(\mathbb{R}^3)} \le M(\mathcal{M}(u_0),\nu)$ uniformly, via a three-step strategy: (i) local weighted L^∞ propagation from Kato mild-solution theory + the Brandolese–Meyer generic-decay framework; (ii) global propagation up to T^* via a Gronwall-type estimate on the weighted quantity $w = (1+|x|)u$; (iii) scale-invariance transfer to the rescaled sequence.

Step (iii) is elementary; step (i) is classical. Step (ii) is the novel mathematical content, and closing it rigorously would resolve the Clay regularity question for Schwartz data under this mildly strengthened hypothesis. A preliminary estimate analysis [7, desk/04] shows that the naive bootstrap argument gives a short-time bound (closed under a quadratic Volterra inequality $\tilde{A} \le L_0 + L_1\tilde{A}^2$) but does not obviously extend to all of $[0,T^*)$; the extension requires either (a) a uniform-in-t Caffarelli–Kohn–Nirenberg ϵ-regularity bound on the exterior of the singular set, giving $|\nabla^k u(x,t)| \le C_k(1+|x|)^{-1-k}$ for $|x| \ge R_0$, or (b) a weighted L^2 energy estimate that propagates $\|(1+|x|)^\alpha u(\cdot,t)\|_{L^2}$ up to T^*.

The full argument, together with the exterior ϵ-regularity analysis, is deferred to a forthcoming work. The present paper establishes Theorem 1.1 unconditionally and records this closure as the open problem reducing the Clay chain.

13. The obstruction landscape

The research program connecting Theorem 1.1 to the bounded-Liouville conjecture for the full regularity problem runs through nine structural obstructions catalogued in the companion corpus [7]. We list them briefly here; full statements, canonical citations, and alternative-route discussions are in the corpus.

(O1) **Closure Obstruction**: no finite combination of the fifty-six antecedent algebraic identities plus Cauchy–Schwarz/Hodge/Helmholtz/coarea closes the Final Sub-Question strain-L^2 improvement.

(O2) **Exchange-of-Limits Obstruction**: the hull topology does not commute with the far-field Rayleigh-quotient limit along translation-hull sequences.

(O3) **Riesz Obstruction**: the pressure Poisson equation plus L^2 bounds does not determine the sign of the pressure Hessian at the eigenfunction supremum.

(O4) **USC Obstruction**: upper semi-continuity of the asymptotic Rayleigh quotient at constants is equivalent to, not independent of, bounded Liouville.

(O5) **Carleman Obstruction**: unique continuation (UCP) is categorically distinct from Liouville (non-existence); the two cannot compose without exponent loss.

(O6) **Integral-Equation Obstruction**: under linear hypotheses, the Birman–Schwinger integral equation admits bounded eigenfunctions at negative eigenvalue; linear spectral analysis is exhausted.

(O7) **Resonance-Diagonal Obstruction**: in the Φ-adic cross-shell coupling matrix on Fibonacci shells, the pressure correction sign is undetermined per shell — a per-scale version of the Riesz obstruction.

(O8) **Helicity-Sector Obstruction**: the symmetric geometric factor $\Lambda_k^{(S)}$ is independent of the scalar quadratic NS invariants; helicity-based closures do not fix its sign.

(O9) **Twist-Writhe Obstruction**: the Călugăreanu–White–Fuller decomposition and the sector decomposition live in incommensurable representations; the $\ell = 2$ angular harmonic on Φ-shells is the irreducible obstruction.

The landscape is *topological, not metric*: each obstruction identifies a signed quantity with $O(C(M))$-controlled magnitude but undetermined sign. Alternative-route research programs — hull-ergodic closure via the Grand Unification Conjecture, icosahedral averaging of the $\ell = 2$ component on Φ-shells, space-time hull closure for Type II — are developed in the companion corpus.

14. Discussion

14.1. **The vortex-stretching reading.** Theorem 1.1 admits a physical reading. The weighted energy $\mathcal{E}(\rho) = \int |\nabla v|^2 \varphi(|x|/\rho) dx$ measures *fluid enstrophy* at scale ρ, and the derivative $\mathcal{E}'(\rho)$ is a local influx rate across the sphere $|x| = \rho$. The superlinear ODE $\mathcal{E}' \geq c\rho^{-\alpha}\mathcal{E}^{\beta}$ with $\beta > 1$ states that the influx rate grows superlinearly with the enstrophy already present: vortex-stretching amplifies strain, which amplifies vorticity, in a feedback loop. The argument is in spirit a no-go: such a feedback, applied across the bulk of $\mathbb{R}^3$, cannot stabilize; the only fixed point is $v \equiv 0$.

The Lorentz hypothesis $v \in L^{3,\infty}(\mathbb{R}^3)$ enters as the sub-scaling-critical bound on the tail of v that keeps the feedback from saturating. At the endpoint $L^{3,\infty}$ the bound is still strong enough, but only barely — the ρ^{γ} factor in the ODE degenerates at $p \to 3$, and vanishes at the intermediate $p = 3/2$. Between these, the structural exponents $\alpha < 1 < \beta$ are strict, and the argument closes.

14.2. **Geometric architecture.** The proof relies on two distinct geometric choices that are sharp:

Polynomial (fixed-ratio) annuli versus exponential weights. The fixed-ratio annulus $A(\theta\rho, \rho)$ with $\theta \in (0, 1)$ fixed is geometrically scale-invariant, which is what allows the Bogovskiĭ operator on $A(\theta\rho, \rho)$ to carry uniform-in-ρ constants (Corollary 4.4). A Gaussian or exponential weight (common in the dynamical arguments of [14, 9]) gives strictly better decay on the tail but destroys scale-invariance of the Bogovskiĭ constant, requiring ρ-dependent bookkeeping that ultimately fails at the critical exponent.

The hat function versus smooth cutoffs. The sharp hat-function φ of Notation 2.1 gives the exact derivative identity (2) with no ϵ-bookkeeping. A smoothed cutoff would require a transition-layer argument that introduces additional surface integrals; these wash out in the limit but obscure the structural form of the inequality.

14.3. **Scope and limitations.** Theorem 1.1 treats the stationary case exclusively; the evolution case is the KNSS composition discussed in §12, where the remaining bottleneck is the uniform $L^{3,\infty}$ inheritance of the rescaled ancient solution. The Liouville step itself is proved unconditionally.

The $n \geq 5$ case is inaccessible by the present method (see Remark 11.1). This is not an artefact: no $L^{n,\infty}$ Liouville theorem is known for stationary NS on $\mathbb{R}^n$ with $n \geq 5$, and the structural obstruction (the Sobolev / integration exponent conflict) suggests that different machinery is needed.

The constants tracked in Corollary 1.4 are polynomial in the initial $L^{3,\infty}$ bound M, which is a quantitative improvement over conjectural triple-exponential bounds ([25]) that would arise from the full-evolution case. However, the constants here are interior-to-the-argument; making them optimal (in the θ, ν dependence) is out of scope.

14.4. **Open problems.** We close with four problems, in decreasing order of specificity.

(P1) **The $L^{3,\infty}$ endpoint.** Resolve Phuc's open problem ([22], reproduced in §12.2): is $\|u\|_{L_t^\infty L_x^{3,\infty}(B_1\times(-1,0))}$ a regularity criterion? Theorem 1.1 is the stationary half; the parabolic analog with backward uniqueness substitutes is the outstanding mathematical question.

(P2) **The weighted-moment proposal.** Close the weighted-moment closure of §12.4 for Schwartz data: prove that $(1+|x|)u_0 \in L^\infty(\mathbb{R}^3)$ implies a uniform $L^{3,\infty}$ bound on the rescaled ancient solution. The main open step is the exterior ϵ-regularity transport to obtain uniform $|\nabla^k u(x,t)| \leq C_k(1+|x|)^{-1-k}$ for $|x|$ large.

(P3) **Non-axisymmetric Nazarov–Ural'tseva.** Find a non-axisymmetric scalar invariant of a 3D NS flow that satisfies a parabolic equation with divergence-free drift amenable to Nazarov–Ural'tseva Harnack [19]. The axisymmetric rv_θ has no known general-case analog; identifying one would generalize [20].

(P4) **Lean formalization.** Formalize Theorem 1.1 in Lean 4/Mathlib. The companion corpus [7] contains a skeleton with three open `sorry`s in the Caccioppoli, ODE contradiction, and Dirichlet-integral-zero steps. Closing those would upgrade the present paper to machine-verified status.

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